

OLIST E-COMMERCE ANALYSIS

KPI & DAX DOCUMENTATION

Project: Olist E-Commerce Analysis
Document Type: KPI & DAX Documentation
Technology: Microsoft Power BI / DAX
Total DAX Measures: 67
KPI Categories: 7
Author: Sejal Sayam
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1. DOCUMENT INFORMATION

Field	Details
Project Name	Olist E-Commerce Analysis
Document Name	KPI & DAX Documentation
Document Type	KPI Documentation
BI Tool	Microsoft Power BI
Formula Language	DAX
Total DAX Measures	67
KPI Categories	7
Version	1.0
Author	Sejal Sayam
Dataset	Brazilian E-Commerce Public Dataset by Olist

2. KPI OVERVIEW

The Olist E-Commerce Analysis Power BI solution contains 67 DAX measures developed to transform raw e-commerce data into meaningful business performance indicators.

The measures are organized into seven business-focused KPI categories.

KPI Category	Number of Measures
Customer KPIs	9
Delivery KPIs	10
Payment KPIs	15
Product KPIs	6
Review KPIs	7
Sales KPIs	14
Seller KPIs	6
Total	67

KPI Framework Objectives

The KPI framework supports analysis of:

- Sales and revenue performance
- Customer acquisition and retention
- Product performance
- Seller performance
- Payment behavior
- Customer satisfaction
- Delivery and logistics performance

3. CUSTOMER KPIs

Customer KPIs are used to understand customer volume, purchasing behavior, repeat purchases, customer acquisition, and geographic revenue contribution.

3.1 Average Orders Per Customer

Definition:

Average number of orders placed per unique customer.

DAX Measure:

Average Orders Per Customer =

`DIVIDE([Total Orders],[Unique Customers])`

Business Purpose:

Measures average customer purchasing frequency and helps evaluate customer engagement.

3.2 Highest Revenue Customer**Definition:**

Identifies the customer who generated the highest revenue.

DAX Measure:

Highest Revenue Customer =

VAR T =

TOPN(

1,

ADDCOLUMNS(

VALUES(customers[customer_unique_id]),

"Revenue", [Total Revenue]

),

[Revenue], DESC

)

RETURN

MAXX(T, customers[customer_unique_id])

Business Purpose:

Helps identify the highest-value customer in the available dataset.

3.3 Highest Spending City**Definition:**

Identifies the customer city generating the highest revenue.

DAX Measure:

Highest Spending City =

VAR T =

TOPN(

```

1,
ADDCOLUMNS(
    VALUES(customers[customer_city]),
    "Revenue", [Total Revenue]
),
[Revenue], DESC
)

```

RETURN

MAXX(T, customers[customer_city])

Business Purpose:

Supports geographic analysis and helps identify high-performing customer locations.

3.4 New Customers

Definition:

Counts distinct customers in the current filter context.

DAX Measure:

New Customers =

DISTINCTCOUNT(customers[customer_id])

Business Purpose:

Used to monitor customer acquisition across the selected analysis period.

Note:

This measure represents distinct customers in the current filter context; it should not be interpreted as first-time customers unless the date/filter logic specifically establishes that definition.

3.5 Previous Month New Customers

Definition:

Calculates the customer count for the previous month using the Calendar table.

DAX Measure:

Previous Month New Customers =

```

CALCULATE(
    [New Customers],

```

```
DATEADD('Calendar'[Date], -1, MONTH)  
)
```

Business Purpose:

Provides a baseline for month-over-month customer comparison.

3.6 Products Per Order

Definition:

Average number of products associated with each order.

DAX Measure:

Products Per Order =

```
DIVIDE([Total Products Sold],[Total Orders])
```

Business Purpose:

Measures average basket size and helps understand purchasing behavior.

3.7 Repeat Customers

Definition:

Counts customers who have placed more than one order.

DAX Measure:

Repeat Customers =

```
CALCULATE(  
    DISTINCTCOUNT(customers[customer_unique_id]),  
    FILTER(  
        VALUES(customers[customer_unique_id]),  
        CALCULATE(COUNTROWS(orders)) > 1  
    )  
)
```

Business Purpose:

Measures repeat purchasing behavior and provides an indication of customer retention.

3.8 Total Customers

Definition:

Counts distinct customer identities using the customer unique ID.

DAX Measure:

Total Customers =

DISTINCTCOUNT(customers[customer_unique_id])

Business Purpose:

Represents the overall customer base available for analysis.

3.9 Unique Customers

Definition:

Counts distinct non-blank customer IDs.

DAX Measure:

Unique Customers =

DISTINCTCOUNTNOBLANK(customers[customer_id])

Business Purpose:

Provides a unique customer count used in customer-related calculations such as revenue per customer.

4. DELIVERY KPIs

Delivery KPIs measure delivery speed, shipping performance, and adherence to estimated delivery dates.

4.1 Average Delivery Days

Definition:

Average number of days between order purchase and customer delivery.

DAX Measure:

Average Delivery Days =

AVERAGEX(

orders,

DATEDIFF(

orders[order_purchase_timestamp],

```

        orders[order_delivered_customer_date],
        DAY
    )
)

```

Business Purpose:

Measures the overall delivery duration experienced by customers.

4.2 Average Delivery Time

Definition:

Average number of days between order purchase and actual customer delivery, excluding orders without a delivery date.

DAX Measure:

Average Delivery Time =

```

AVERAGEX(
    FILTER(
        orders,
        NOT(ISBLANK(orders[order_delivered_customer_date]))
    ),
    DATEDIFF(
        orders[order_purchase_timestamp],
        orders[order_delivered_customer_date],
        DAY
    )
)

```

Business Purpose:

Provides a reliable measure of actual customer delivery performance.

4.3 Average Shipping Days

Definition:

Average number of days between order purchase and delivery to the carrier.

DAX Measure:

Average Shipping Days =

```
AVERAGEX(  
    FILTER(  
        orders,  
        NOT(ISBLANK(orders[order_delivered_carrier_date]))  
    ),  
    DATEDIFF(  
        orders[order_purchase_timestamp],  
        orders[order_delivered_carrier_date],  
        DAY  
    )  
)
```

Business Purpose:

Helps evaluate how quickly orders move from purchase to carrier handoff.

4.4 Late Deliveries**Definition:**

Counts orders delivered after the estimated delivery date.

DAX Measure:

Late Deliveries =

```
COUNTROWS(  
    FILTER(  
        orders,  
        orders[order_delivered_customer_date] >  
        orders[order_estimated_delivery_date]  
    )  
)
```


Business Purpose:

Identifies the volume of delayed deliveries and supports logistics performance monitoring.

4.5 Maximum**Definition:**

Maximum threshold value used in the delivery dashboard.

DAX Measure:

Maximum = 10

Business Purpose:

Used as a dashboard visualization/reference threshold.

4.6 Minimum**Definition:**

Minimum threshold value used in the delivery dashboard.

DAX Measure:

Minimum = 0

Business Purpose:

Used as a dashboard visualization/reference threshold.

4.7 On-Time Deliveries**Definition:**

Counts orders delivered on or before the estimated delivery date.

DAX Measure:

On-Time Deliveries =

```
COUNTROWS(  
    FILTER(  
        orders,  
        orders[order_delivered_customer_date] <=  
        orders[order_estimated_delivery_date]  
    )  
)
```

)

Business Purpose:

Measures the volume of orders delivered within the expected timeframe.

4.8 On-Time Delivery %

Definition:

Percentage of total orders that were delivered on or before the estimated delivery date.

DAX Measure:

On-Time Delivery % =

DIVIDE(

[On-Time Deliveries],

[Total Orders],

0

)

Business Purpose:

Provides a high-level measure of delivery reliability.

4.9 Shipping Target

Definition:

Reference target value used for shipping performance analysis.

DAX Measure:

Shipping Target = 3

Business Purpose:

Provides a benchmark for comparing shipping duration against the defined target.

4.10 Target

Definition:

Reference target used for delivery performance visualization.

DAX Measure:

Target = 0.95

Business Purpose:

Represents a 95% reference target for delivery performance.

5. PAYMENT KPIs

Payment KPIs analyze payment value, transaction volume, installment behavior, and payment method usage.

5.1 Average Installments**Definition:**

Average number of installments used across payment transactions.

DAX Measure:

Average Installments =

AVERAGE(payments[payment_installments])

Business Purpose:

Helps understand customer installment preferences.

5.2 Average Payment**Definition:**

Average payment value per payment transaction.

DAX Measure:

Average Payment =

AVERAGE(payments[payment_value])

Business Purpose:

Measures the average monetary value of payment transactions.

5.3 Boleto Payments**Definition:**

Total payment value made using the boleto payment method.

DAX Measure:

Boleto Payments =

CALCULATE(

```
SUM(payments[payment_value]),  
payments[payment_type] = "boleto"  
)
```

Business Purpose:

Measures payment value generated through boleto.

5.4 Credit Card Orders

Definition:

Counts distinct orders that used credit card payment.

DAX Measure:

Credit Card Orders =

```
CALCULATE(  
    DISTINCTCOUNT(payments[order_id]),  
    payments[payment_type] = "credit_card"  
)
```

Business Purpose:

Measures customer adoption of credit card payments.

5.5 Credit Card Payments

Definition:

Total payment value made using credit cards.

DAX Measure:

Credit Card Payments =

```
CALCULATE(  
    SUM(payments[payment_value]),  
    payments[payment_type] = "credit_card"  
)
```

Business Purpose:

Measures revenue/payment value associated with credit card transactions.

5.6 Debit Card Payments

Definition:

Total payment value made using debit cards.

DAX Measure:

Debit Card Payments =

```
CALCULATE(  
    SUM(payments[payment_value]),  
    payments[payment_type] = "debit_card"  
)
```

Business Purpose:

Measures payment value generated through debit cards.

5.7 Highest Installment

Definition:

Highest number of installments recorded.

DAX Measure:

Highest Installment =

```
MAX(payments[payment_installments])
```

Business Purpose:

Identifies the maximum installment plan used by customers.

5.8 Lowest Installment

Definition:

Lowest number of installments recorded.

DAX Measure:

Lowest Installment =

```
MIN(payments[payment_installments])
```

Business Purpose:

Identifies the minimum installment value recorded.

5.9 Maximum Payment

Definition:

Highest individual payment value recorded.

DAX Measure:

Maximum Payment =

MAX(payments[payment_value])

Business Purpose:

Identifies the largest payment transaction.

5.10 Minimum Payment

Definition:

Lowest individual payment value recorded.

DAX Measure:

Minimum Payment =

MIN(payments[payment_value])

Business Purpose:

Identifies the smallest payment transaction.

5.11 Payment Per Order

Definition:

Average payment value per distinct order.

DAX Measure:

Payment per Order =

```
DIVIDE(  
    [Total Payment],  
    DISTINCTCOUNT(payments[order_id])  
)
```

Business Purpose:

Measures the average payment amount associated with each order.

5.12 Total Installments

Definition:

Total number of installments across all payment transactions.

DAX Measure:

Total Installments =

SUM(payments[payment_installments])

Business Purpose:

Provides an overall view of installment usage.

5.13 Total Payment

Definition:

Total payment value recorded across all payment transactions.

DAX Measure:

Total Payment =

SUM(payments[payment_value])

Business Purpose:

Measures the total payment value processed in the dataset.

5.14 Total Transactions

Definition:

Total number of payment records.

DAX Measure:

Total Transactions =

COUNTROWS(payments)

Business Purpose:

Measures overall payment transaction volume.

5.15 Voucher Payments

Definition:

Total payment value made using vouchers.

DAX Measure:

Voucher Payments =

```
CALCULATE(  
    SUM(payments[payment_value]),  
    payments[payment_type] = "voucher"  
)
```

Business Purpose:

Measures the value of transactions associated with voucher payments.

6. PRODUCT KPIs

Product KPIs measure product volume, pricing, freight, weight, and category performance.

6.1 Average Freight Per Product

Definition:

Average freight value per product sold.

DAX Measure:

Average Freight Per Product =

`DIVIDE([Total Freight],[Total Products Sold])`

Business Purpose:

Helps evaluate average logistics cost associated with product sales.

6.2 Average Product Weight

Definition:

Average weight of products in the product catalog.

DAX Measure:

Average Product Weight =

`AVERAGE(products[product_weight_g])`

Business Purpose:

Supports logistics and product profile analysis.

6.3 Revenue Per Product

Definition:

Average revenue generated per unique product.

DAX Measure:

Revenue Per Product =

`DIVIDE([Total Revenue],[Unique Products])`

Business Purpose:

Helps evaluate revenue contribution relative to the product catalog.

6.4 Top Selling Category

Definition:

Identifies the product category with the highest product sales value based on item price.

DAX Measure:

Top Selling Category =

```
MAXX(  
    TOPN(  
        1,  
        VALUES(products[product_category_name]),  
        CALCULATE(SUM(items[price])),  
        DESC  
    ),  
    products[product_category_name]  
)
```

Business Purpose:

Identifies the strongest-performing product category.

6.5 Total Products Sold

Definition:

Total number of product-item records sold.

DAX Measure:

Total Products Sold =

COUNTROWS(items)

Business Purpose:

Measures overall product sales volume.

6.6 Unique Products

Definition:

Number of distinct products sold or represented in the product data.

DAX Measure:

Unique Products =

DISTINCTCOUNT(products[product_id])

Business Purpose:

Measures product assortment and supports product-level analysis.

7. REVIEW KPIs

Review KPIs measure customer satisfaction and review distribution.

7.1 Average Review Score

Definition:

Average customer review score.

DAX Measure:

Average Review Score =

AVERAGE(review[review_score])

Business Purpose:

Provides an overall indicator of customer satisfaction.

7.2 Maximum Review Score

Definition:

Highest review score recorded.

DAX Measure:

Maximum Review Score =

MAX(review[review_score])

Business Purpose:

Identifies the highest customer rating present in the dataset.

7.3 Minimum Review Score

Definition:

Lowest review score recorded.

DAX Measure:

Minimum Review Score =

MIN(review[review_score])

Business Purpose:

Identifies the lowest customer rating present in the dataset.

7.4 Negative Reviews

Definition:

Number of reviews with a score of 1 or 2.

DAX Measure:

Negative Reviews =

```
CALCULATE(  
    COUNTROWS(review),  
    review[review_score] <= 2  
)
```

Business Purpose:

Helps identify customer dissatisfaction.

7.5 Neutral Reviews

Definition:

Number of reviews with a score of 3.

DAX Measure:

Neutral Reviews =

```
CALCULATE(  
    COUNTROWS(review),  
    review[review_score] = 3  
)
```

Business Purpose:

Measures neutral customer feedback.

7.6 Positive Reviews

Definition:

Number of reviews with a score of 4 or 5.

DAX Measure:

Positive Reviews =

```
CALCULATE(  
    COUNTROWS(review),  
    review[review_score] >= 4  
)
```

Business Purpose:

Measures positive customer sentiment.

7.7 Total Reviews

Definition:

Total number of review records.

DAX Measure:

Total Reviews =

```
COUNT(review[review_id])
```

Business Purpose:

Measures the volume of customer feedback available for analysis.

8. SALES KPIs

Sales KPIs measure revenue, order volume, growth, product pricing, freight, and customer revenue contribution.

8.1 Average Daily Revenue

Definition:

Average revenue generated per day in the selected period.

DAX Measure:

Average Daily Revenue =

```
AVERAGEX(  
    VALUES(Calendar[Date]),  
    [Total Revenue]  
)
```

Business Purpose:

Measures average daily sales performance.

8.2 Average Order Value

Definition:

Average revenue generated per order.

DAX Measure:

Average Order Value =

```
DIVIDE([Total Revenue],[Total Orders],0)
```

Business Purpose:

Measures the average monetary value of an order.

8.3 Average Product Price

Definition:

Average price of products sold.

DAX Measure:

Average Product Price =

AVERAGE(items[price])

Business Purpose:

Helps understand the average selling price of products.

8.4 Gross Sales

Definition:

Total revenue plus total freight value based on the project's calculation.

DAX Measure:

Gross Sales =

[Total Revenue] + [Total Freight]

Business Purpose:

Provides a combined view of product revenue and freight value.

8.5 Monthly Growth %

Definition:

Month-over-month percentage change in new customer count.

DAX Measure:

Monthly Growth % =

VAR Curr = [New Customers]

VAR Prev = [Previous Month New Customers]

RETURN

IF(
 ISBLANK(Prev),
 BLANK(),
 DIVIDE(Curr - Prev, Prev)
)

Business Purpose:

Measures monthly change in customer acquisition.

Note:

Despite being located under Sales KPIs, this measure specifically calculates **new customer growth**, not revenue growth.

8.6 Previous Year Revenue

Definition:

Revenue generated during the corresponding period in the previous year.

DAX Measure:

Previous Year Revenue =

```
CALCULATE(  
    [Total Revenue],  
    SAMEPERIODLASTYEAR(Calendar[Date])  
)
```

Business Purpose:

Provides a baseline for year-over-year revenue comparison.

8.7 Profit

Definition:

Analytical profit proxy calculated as product price minus freight value.

DAX Measure:

Profit =

```
SUM(items[price]) - SUM(items[freight_value])
```

Business Purpose:

Provides an analytical profitability indicator based on the available price and freight fields.

Important Note:

This should be treated as an **estimated analytical profit measure**, not accounting profit, because the dataset does not provide a complete business cost structure.

8.8 Revenue Per Customer

Definition:

Average revenue generated per unique customer.

DAX Measure:

Revenue Per Customer =

DIVIDE([Total Revenue],[Unique Customers])

Business Purpose:

Measures average customer revenue contribution.

8.9 Revenue Per Order

Definition:

Average revenue generated per order.

DAX Measure:

Revenue Per Order =

DIVIDE([Total Revenue],[Total Orders])

Business Purpose:

Measures average revenue contribution per order.

8.10 Running Total Revenue

Definition:

Cumulative revenue from the beginning of the available calendar period through the selected date.

DAX Measure:

Running Total Revenue =

```
CALCULATE(  
    [Total Revenue],  
    FILTER(  
        ALL(Calendar[Date]),  
        Calendar[Date] <= MAX(Calendar[Date])  
    )  
)
```


)

Business Purpose:

Helps analyze cumulative revenue growth over time.

8.11 Total Freight

Definition:

Total freight value associated with product items.

DAX Measure:

Total Freight =

SUM(items[freight_value])

Business Purpose:

Measures the overall freight value associated with sales.

8.12 Total Orders

Definition:

Number of distinct orders.

DAX Measure:

Total Orders =

DISTINCTCOUNT(orders[order_id])

Business Purpose:

Measures overall order volume.

8.13 Total Revenue

Definition:

Total product sales revenue based on item price.

DAX Measure:

Total Revenue =

SUM(items[price])

Business Purpose:

Serves as the primary revenue KPI used throughout the Power BI analysis.

8.14 YoY Growth %

Definition:

Percentage change in revenue compared with the corresponding previous-year period.

DAX Measure:

YoY Growth % =

```
DIVIDE(  
    [Total Revenue] - [Previous Year Revenue],  
    [Previous Year Revenue],  
    0  
)
```

Business Purpose:

Measures year-over-year revenue growth or decline.

9. SELLER KPIs

Seller KPIs evaluate seller activity, order fulfillment, revenue contribution, and seller concentration.

9.1 Cancelled Orders

Definition:

Number of distinct orders with a cancelled status.

DAX Measure:

Cancelled Orders =

```
CALCULATE(  
    DISTINCTCOUNT(orders[order_id]),  
    orders[order_status] = "canceled"  
)
```

Business Purpose:

Measures order cancellation volume.

9.2 Delivered Orders

Definition:

Number of distinct orders with a delivered status.

DAX Measure:

Delivered Orders =

```
CALCULATE(  
    DISTINCTCOUNT(orders[order_id]),  
    orders[order_status] = "delivered"  
)
```

Business Purpose:

Measures successfully completed order volume.

9.3 Orders Per Seller

Definition:

Average number of orders associated with each unique seller.

DAX Measure:

Orders Per Seller =

```
DIVIDE([Total Orders],[Unique Sellers])
```

Business Purpose:

Measures average seller order volume.

9.4 Revenue Per Seller

Definition:

Average revenue generated per unique seller.

DAX Measure:

Revenue Per Seller =

```
DIVIDE([Total Revenue],[Unique Sellers])
```

Business Purpose:

Measures average seller revenue contribution.

9.5 Top Seller Revenue

Definition:

Highest revenue generated by an individual seller.

DAX Measure:

Top Seller Revenue =

```
MAXX(  
    VALUES(items[seller_id]),  
    CALCULATE(SUM(items[price]))  
)
```

Business Purpose:

Identifies the highest-revenue seller and supports seller performance analysis.

9.6 Unique Sellers

Definition:

Number of distinct sellers.

DAX Measure:

Unique Sellers =

```
DISTINCTCOUNT(sellers[seller_id])
```

Business Purpose:

Measures the size of the seller base.

10. KPI SUMMARY

The Olist E-Commerce Analysis Power BI model contains **67 DAX measures** designed to provide a comprehensive analytical view of the e-commerce business.

The KPI framework covers seven key business areas:

Area	Business Focus
Customer	Acquisition, retention, purchasing behavior
Delivery	Delivery speed and reliability

Area	Business Focus
Payment	Payment value, transactions and methods
Product	Product sales, pricing and freight
Review	Customer satisfaction and sentiment
Sales	Revenue, orders, growth and profitability proxy
Seller	Seller performance and order fulfillment

These KPIs support the project's five Power BI dashboard areas and enable users to analyze business performance from multiple perspectives.

11. KPI DOCUMENTATION NOTES

Data Context

All KPIs are calculated from the Olist e-commerce dataset and depend on the underlying data model, relationships, calendar table, and filter context used in Power BI.

Filter Context

Most DAX measures are dynamic and change according to filters and slicers applied in the Power BI report.

Revenue Definition

For this project:

Total Revenue = $\text{SUM}(\text{items}[\text{price}])$

Therefore, revenue represents the total item price used in the project's analytical model.

Profit Definition

The project's Profit measure is:

Profit = Total Item Price - Total Freight

It should be interpreted as an **analytical profitability proxy**, not formal accounting profit.

Growth Definition

- **Monthly Growth %** measures month-over-month change in new customers.
- **YoY Growth %** measures year-over-year change in total revenue.

END OF DOCUMENT

Final filename

Save the Word document as:

Olist_KPI_Documentation.docx

Then **do not convert it to PDF yet.**

First, copy all of the above into Word, apply your existing project-documentation formatting, add page numbers, and create a clickable Table of Contents if you want consistency with your main documentation.